

Automated 3D Stenosis Quantification and Visualization Workflow for Coronary Artery Disease Clinical Decision Support

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SANT PAU
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Hospital de
la Santa Creu i
Sant Pau

4 User Experience and Interface Usability Evaluation

Results after the users complete the interaction and the formulary:

- <https://docs.google.com/forms/d/e/1FAIpQLSeHmCs-B8pYm0gK3JLsp3BxIJqsrcMEDv3arWlycSDIPj6ARA/viewform?usp=publish-editor>

4.1 System Usability Score

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4.2 Feedback, Discussions and Future Work with Doctors at HSP

During the final stages of this project, a clinical evaluation session was conducted with expert cardiologists at Hospital de la Santa Creu i Sant Pau to assess the pipeline's clinical utility and identify areas for future improvement. The feedback gathered is highly valuable, as it bridges the gap between technical implementation and real-world clinical workflows. The following sections outline the key discussions and proposed future work.

4.2.1 Algorithmic and Metric Refinements

- **Dynamic Reference Values for Stenosis Calculation:** The current formula uses fixed cross-sectional area reference values depending on a predefined window size, which introduces limitations in quantification, particularly near bifurcations. Clinical experts noted that they typically select healthy reference points within the same segment manually. Future iterations should allow clinicians to dynamically adjust the proximal and distal reference values, bypassing anatomical anomalies and improving the accuracy of the stenosis calculation.
- **Centerline Resolution Optimization:** The current pipeline extracts centerline points with high spatial granularity (< 0.1 mm for short branches, < 0.2 mm for long branches). While this provides rich geometric data, cardiologists suggested that a coarser spacing (e.g., ~ 2 mm) is generally sufficient for standard diagnosis. Reducing the resolution would significantly improve computational efficiency, though higher granularity could be dynamically retained exclusively in diseased segments.
- **Plaque vs. Stenosis Quantification:** A critical clinical distinction was emphasized during the evaluation: while plaque accumulation (e.g., calcium, cholesterol) causes stenosis, the presence of plaque does not always imply a hemodynamically significant stenosis. Current clinical workflows prioritize identifying plaque first. Future work should integrate plaque quantification alongside stenosis detection, providing a more comprehensive and realistic diagnostic picture.

4.2.2 Visualization and Interface Enhancements

- **Continuous Lumen Visualization:** Clinicians expressed the need to visualize the lumen's progression continuously along the artery, rather than relying solely on discrete centerline data points. Providing continuous cross-sectional profiles allows for an easier and more intuitive diagnostic assessment.
- **Integration of Medical Imaging (Ground Truth):** A critical requirement for clinical adoption is the ability to compare algorithmic results with raw medical images. Future

versions of the dashboard must incorporate Curved Multi-Planar Reformation (CMPR) and cross-sectional views of the real Coronary Computed Tomography Angiography (CCTA) scans. This overlay ensures clinicians can validate the software's findings against the anatomical reality.

- **Advanced Dashboard Features and Explainability:** Taking inspiration from robust commercial solutions, the ultimate goal for the user interface is a highly interactive dashboard. Key requested components include a 3D representation of the coronary tree, interactive metrics where doctors can adjust thresholds, and clear explainability of the algorithmic decisions to avoid "black-box" models. Furthermore, rather than displaying per-point stenosis metrics, the interface should highlight aggregated affected zones (e.g., regions with $\geq 50\%$ stenosis) to allow doctors to quickly triage or inspect specific areas.

4.2.3 Clinical Workflow Integration

The most significant potential of this visualization tool lies in its integration within a broader hospital ecosystem. By unifying automatic segmentation inputs with advanced patient prioritization algorithms, this project serves as a crucial connective element. In a clinical setting, a doctor reviewing a prioritized list of at-risk patients could open a pop-up of this dashboard for a rapid, initial assessment. This quick overview of key metrics and affected zones would allow clinicians to efficiently decide whether a patient requires immediate, detailed analysis using complex diagnostic software or if they can be safely managed otherwise.

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